

# Confidence, not conflict: which value $\rightarrow$ RT mapping best describes probability discounting, and does RT help?

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| Name                                                                                                                                                                 | Matriculation no. | Contribution                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------|
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| The valuation model, choice rule, comparison framework and report were developed jointly; the three authors contributed equally. Requested assessment: full grading. |                   |                                            |

**Abstract** We model the choices and the reaction times of 49 participants in a probability discounting task with reward and loss conditions. One valuation model is held fixed: a Rachlin hyperboloid in the odds against reward,  $V_i = r_i / (1 + h\theta_i)$  applied to both options, with separate rates for gains and losses, coupled to choice by a logistic rule on the normalised value difference  $\Delta V$ . Three RT models sit on top of it and differ only in how  $\Delta V$  enters the location of a shifted log-normal: linear conflict ( $|\Delta V|$ ), decision uncertainty ( $|2P - 1|$ ) and quadratic conflict ( $\Delta V^2$ ). A structural feature of the data governs everything that follows: run A is sure-vs-risky and run B risky-vs-risky, so both options must be discounted and run B is an unseen decision structure rather than a retest. Estimation is per participant by MAP on run A. On the quantity that separates the mappings, the run B RT predictive log-likelihood, decision uncertainty wins at  $-1.492$  nats per trial against  $-1.536$  and  $-1.562$  ( $p = 0.040, 0.004$ ), and it is the one ranking here that replicates from in-sample. Against a choice-only baseline fitted to the identical model, adding RT changes held-out choice prediction by  $-0.037$ ,  $+0.003$  and  $+0.025$  nats per trial, none significant, and it does not stabilise the constructs: every cross-context reliability of the discounting parameters sits near zero and the two shifts excluding zero are losses. A same-task control locates the cause: fitting half of run A and predicting the other half gives  $-0.464$  nats per trial with 49 of 49 participants above chance, against  $-1.004$  and 6 of 49 across tasks. We recommend decision uncertainty as the best description of response time, while concluding that modelling RT does not improve the account of the choices. Non-decision time is not identifiable.

## 1 INTRODUCTION

Every trial of a value-based choice task yields two observations: which option was taken, and how long the participant took to take it. Discounting research almost always uses only the first, because discount rates are traditionally estimated from titrated indifference points, which summarise a block of choices and discard their timing, and because discounting models are static value functions with no dimension along which a response time could be predicted (Green & Myerson, 2004). Both modalities are nevertheless driven by the same latent quantity: if subjective value drives the choice, it should also govern how long the comparison takes, making RT a second and essentially free measurement of the same construct. If RTs carry information about  $h$  and  $\beta$ , fitting both should sharpen those estimates; if not, the joint fit pays for the RT term and gets nothing back.

We therefore hold valuation and choice entirely fixed, using a Rachlin hyperboloid in odds against (Rachlin et al., 1991) with a logistic choice rule, and vary only the value  $\rightarrow$  RT mapping: linear conflict (M1), the first-order behaviour of an accumulator with drift proportional to  $\Delta V$  (Ratcliff, 1978; Ratcliff & McKoon, 2008); decision uncertainty (M2), which saturates once a participant is confident; and quadratic conflict (M3), which concentrates the slowing near indifference. The three span the plausible curvatures, and the report asks in turn which mapping predicts RT best out of sample, whether its parameters are identifiable and reliable, and whether modelling RT improves the account of the choices.

## 2 METHODS

### 2.1 Data

The probability discounting data set contains 49 participants, each with an inference run (A) and an evaluation run (B) of 200 trials. On each trial a smaller reward  $r_{cert}$  is offered against a larger reward  $r_{unc}$  paid with

probability  $p \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ , which we summarise by the odds against,  $\theta = (1 - p)/p$ . A reward condition (both magnitudes positive) and a loss condition (both negative, verified in the raw files, so no sign correction is applied anywhere) are fully crossed with the design at 100 trials per condition per run.

*The two runs are not the same task.* `p_cert` is absent for every run A trial and present for every run B trial (0.3–0.7), in all 49 participants: run A is sure-vs-risky ( $\theta_{cert} = 0$ ), run B risky-vs-risky ( $\theta_{cert} \in [0.43, 2.33]$ ). The `cert/unc` labels are inherited from run A, but what the runs share is magnitude, not certainty. Section 2.2 discounts both options with one equation, and the runs are read as two decision contexts, not two administrations of one task.

We drop 68 non-responses (0.35%) and 105 trials faster than 250 ms (0.54%), a fixed global floor for anticipatory responses, leaving 19,427 of 19,600. No upper cut-off is applied: neither extreme (200 ms, 9996 ms) shows a pile-up, so there is no sign of censoring. Four participants make more than 10% of their responses below 400 ms; we flag and keep them. Participants choose the uncertain option on 46.4% of reward and 43.2% of loss trials (between-participant SDs 0.14, 0.10), respond more slowly under losses (median 2.09 s) than gains (1.81 s), and differ enormously in speed (participant median RTs span 0.40–3.50 s). Raw RT is right-skewed (1.55) and near-symmetric on the log scale (−0.51), which is what recommends a log-normal observation model.

Before fitting we check the valuation model model-free. A logistic of  $\log(r_{unc}/r_{cert})$  fitted at each  $p$  and condition gives the indifference ratio at that  $\theta$ , and a hyperboloid predicts  $r_{unc}/r_{cert} = 1 + h\theta$ , linear in  $\theta$ ; the rearrangement presupposes an undiscounted smaller option, so it is estimated on run A only. Figure 1 shows that it holds, with slopes  $h \approx 3.72$  for gains and  $h \approx 0.64$  for losses, the model-free signature of the reflection effect (Kahneman & Tversky, 1979; Green & Myerson, 2004), and close to the fitted medians below, which is the check that the likelihood is coded correctly.

The runs also differ in coverage and behaviour: run B’s offers move to larger magnitude ratios (mean log ratio 1.73 against 1.32), 23% of its trials fall outside the participant’s own run A range, the uncertain option is chosen more often (0.394 to 0.503,  $p = 0.001$ ; a residual +0.038 survives matching participant, condition,  $p$  and ratio octile), and responses speed up sharply (median 2.19 s to 1.77 s, 45 of 49 faster) while Section 2.3 fixes  $\tau$  and  $a$  from run A. Section 3.2 therefore measures generalisation across decision contexts rather than test–retest, and Section 3.3 extrapolation to an unseen structure; Section 2.4 adds two controls that keep those readings separable.

## 2.2 Models

**Valuation and choice (held fixed).** Both options are discounted by the odds against their own outcome (Rachlin et al., 1991):

$$V_i = \frac{r_i}{1 + h\theta_i}, \quad \theta_i = \frac{1 - p_i}{p_i}, \quad h \geq 0, \quad (1)$$

with separate rates  $h_{gain}$  and  $h_{loss}$  per participant. One equation covers both runs, reducing exactly to  $V_{cert} = r_{cert}$  where  $\theta_{cert} = 0$ ; hard-coding that instead would overstate the smaller option by a factor  $1 + h\theta_{cert}$ , between 1.3 and 8.6 at the fitted rates, on every one of the 9,700 retained run B trials. Because loss magnitudes are stored negative, the same equation produces both risk attitudes at no extra parameter cost:  $V_{unc}$  shrinks towards zero, worse than a sure gain (risk aversion) but better than a sure loss (risk seeking), so the reflection effect emerges for free. Magnitudes span five orders of magnitude, so we normalise,  $\Delta V = (V_{unc} - V_{cert}) / [(|r_{cert}| + |r_{unc}|)/2]$ , which lets one sensitivity serve small and large trials but removes absolute magnitude effects. Choice is logistic,  $P_t \equiv P(\text{uncertain}) = \sigma(\beta \Delta V_t + \beta_0)$ , where  $\beta_0$  absorbs a value-independent bias, giving the Bernoulli trial likelihood  $\log p(y_t | \theta) = y_t \log P_t + (1 - y_t) \log(1 - P_t)$  with  $y_t = 1$  when the uncertain option was chosen.

**RT distribution (held fixed).** RTs follow a shifted log-normal,  $RT = \tau + \exp(\mu + \sigma\varepsilon)$  with  $\varepsilon \sim \mathcal{N}(0, 1)$ , so for  $rt > \tau$

$$\log p(rt | \mu, \sigma, \tau) = -\log(rt - \tau) - \log \sigma - \frac{1}{2} \log 2\pi - \frac{1}{2} \left[ \frac{\log(rt - \tau) - \mu}{\sigma} \right]^2. \quad (2)$$

Holding this distribution identical across the three RT models is what makes the comparison one of mappings rather than of distributional shape. The full trial likelihood is  $p(\text{choice}, RT | \theta) = p(y_t | \theta) p(rt_t | \theta)$ , since the modalities are conditionally independent given the shared latent  $\Delta V$ ; that shared valuation is what makes the model multimodal. The models differ only in the location: M1,  $\mu_t = a - b|\Delta V_t|$ ; M2,  $\mu_t = a - b|2P_t - 1|$ ; M3,  $\mu_t = a - b\Delta V_t^2$ .

**Psychological hypotheses.** M1 says what slows a participant is the distance between the options in value space, every extra unit of advantage buying the same speed, the first-order behaviour of an accumulator with drift proportional to  $\Delta V$  (Ratcliff & McKoon, 2008; Krajbich et al., 2010). M2 says the driver is subjective indecision: its predictor is 0 under complete indecision and 1 under certainty, so once  $P \approx 0.95$  further advantage buys no more speed. M1 has RT tracking value, M2 has it tracking confidence. M3 says deliberation is costly only very near indifference, so the marginal speed-up grows rather than shrinks with  $|\Delta V|$ . The psychological claim in each case is the curvature.

**Choice-only baseline.** The baseline deletes the RT term from the likelihood and fits the identical valuation model and choice rule to the choices alone: not a fourth model but the same model fitted to one modality, and the reference against which the RT modality is valued.

## 2.3 Inference

We estimate parameters per participant by maximum a posteriori on run A, leaving run B untouched at this stage:

$$\log p(\theta \mid \text{data}) \propto \underbrace{\sum_t [y_t \log P_t + (1 - y_t) \log(1 - P_t)]}_{\text{choice}} + \underbrace{\sum_t \log p(rt_t \mid \mu_t, \sigma, \tau)}_{\text{RT; omitted in the baseline}} + \log p(\theta). \quad (3)$$

Four choice parameters are fitted unconstrained on transformed scales (`log_h_gain`, `log_h_loss`, `log_beta`, `beta0`), and the RT models add `a`, `b`, `log_sigma` and `tau_raw`, where  $\tau = \sigma(\text{tau\_raw}) \cdot 0.99 \min(RT)$  enforces  $0 < \tau < \min(RT)$ . Priors are weakly informative Gaussians on the fitted scale (Table 1), wide enough that the data dominate wherever the likelihood has curvature;  $b$ 's is centred at zero, so a positive difficulty effect is never assumed, and is widened to  $\mathcal{N}(0, 3^2)$  for M3, whose predictor is on a smaller scale (median 0.08 against 0.28), which equalises regularisation in effect on  $\mu$  rather than in raw coefficient.

Optimisation uses L-BFGS-B with box constraints and 10 restarts per participant, one from the model-free estimates and the rest from the priors; seeds derive deterministically from (model, run, participant), so the pipeline reproduces exactly. Convergence is reported for every run A fit and every run B fit except M2 (98%). Two diagnostics guard the estimates (Fig. 2): refitting by MLE returns essentially identical parameters for well-identified participants ( $r \geq 0.995$ ), so the priors are not doing the work; and a curvature check flags 19 of 49 participants as poorly identified in at least one parameter (14 in  $\log h_{\text{gain}}$ , 14 in  $\log h_{\text{loss}}$ ), which we keep and track separately.

## 2.4 Evaluation Plan

*Recovery.* For each RT model we draw 200 parameter sets covering the run A estimates, simulate choices and RTs over a real participant's design, and refit twice, with the joint and with the choice-only likelihood; correlation and RMSE quantify recovery, and the error correlations diagnose identifiability. The study is then repeated over run B designs, to ask whether  $h$  is recoverable *in principle* where  $\theta_{\text{cert}} > 0$ , separately from whether it transfers. *Reliability.* The same model is estimated independently on runs A and B and the estimates correlated across participants (Pearson, Spearman, ICC(2,1) (Shrout & Fleiss, 1979)) with 2000-sample bootstrap CIs; CIs for differences apply one participant resample to both models, preserving the pairing. *Prediction.* Run A parameters are applied unchanged to run B, nothing refitted; for choice we report predictive log-likelihood per trial, accuracy, balanced accuracy and AUC, for RT predictive log-likelihood per trial and RMSE on  $\log(RT - \tau)$ , each for all trials and by condition. *Which quantity compares what.* A joint and a choice-only likelihood describe different data, so their totals are not comparable: RT models are compared with each other on the RT predictive log-likelihood only, RT-informed fits with the baseline on choice quantities only, by paired Wilcoxon signed-rank tests. *Same-task control.* Because run B is a different task, a weak run A  $\rightarrow$  run B result is ambiguous between a poor model and a limit on extrapolation, so we also fit each participant's choice model on half of run A and predict the other half: held-out trials, same estimator, same structure on both sides.

### 3 RESULTS

The fits agree with the model-free analysis. Median run A baseline estimates are  $h_{gain} = 2.65$  (IQR 1.00–7.99),  $h_{loss} = 0.82$  (0.21–1.98),  $\beta = 4.53$  and  $\beta_0 = -0.24$ , with  $h_{gain} > h_{loss}$  in 32 of 49 participants: the reflection effect is recovered by the fit as well as by the raw indifference points, and both rates land near the model-free 3.72 and 0.64.

#### 3.1 Parameter Recovery and Identifiability

Every parameter but one recovers well under every RT model, with  $r$  between 0.83 and 0.98 (Fig. 3, Table 2). Non-decision time is the exception.  $\tau$  is not identifiable, with  $r = 0.19$  (M1), 0.21 (M2) and 0.29 (M3) and an upward bias of about +0.05 s, and the recovery errors show why:  $\text{corr}(\text{err}_\tau, \text{err}_a) = -0.64$  under M1 and M3. Both shift the predicted RT distribution and at  $\approx 200$  trials the data cannot separate them, so an over-estimated shift is paid for by an under-estimated intercept. Nothing else trades off appreciably (off-diagonal  $|r| \leq 0.31$ , Fig. 4).

Adding RT improves recovery of the choice parameters in every case, by amounts that differ enormously between mappings. As an RMSE ratio (joint over choice-only, below 1 meaning RT helps), M3 gives 0.47 for  $\log h_{gain}$  and 0.49 for  $\log h_{loss}$ , roughly halving the error, against 0.86 and 0.72 for M1 and 0.98 and 0.88 for M2; paired Wilcoxon tests on absolute errors confirm the improvement under M1 and M3 ( $p < 10^{-4}$ ) and only weakly under M2 ( $p = 0.03$  and 0.16).

Repeating the study over run B designs answers a prior question that Section 3.3 cannot: is  $h$  recoverable at all where  $\theta_{cert} > 0$ ? It is, though less sharply:  $\log h_{gain}$  falls from 0.925/0.900/0.977 to 0.888/0.816/0.931 with RMSE up by roughly 40%, as expected when the identifying signal is split across two entangled discount terms instead of one, and M3’s advantage survives (RMSE ratio 0.61 against 0.79 and 1.01). Whatever fails between the runs in Section 3.2, it is not identifiability.

#### 3.2 Reliability

Because run A and run B are different decision structures, this section measures whether a construct generalises across contexts, not classical test–retest, and on that reading the discounting parameters do not generalise (Fig. 5, Table 3): under the baseline  $\log h_{gain}$  reaches only  $r = 0.180$  (CI  $[-0.08, +0.45]$ , ICC 0.14),  $\log h_{loss}$  0.191,  $\log \beta$  0.215 and  $\beta_0$  0.103, every CI including zero. Such values are low even by the standards of cognitive-task parameters (Hedge et al., 2018; Enkavi et al., 2019); Section 3.3 shows why.

What *is* stable is everything independent of the valuation: the RT level  $a$ ,  $\tau$  and  $\log \sigma$  (for which we report Spearman, since a few extreme-RT participants inflate Pearson). The difficulty slope  $b$  is not stable at all. Participants carry a reproducible speed and spread between the two tasks, but neither a reproducible discount rate nor a reproducible sensitivity of RT to difficulty.

Adding RT does not repair this, and twice makes it worse. Relative to the baseline,  $\Delta r$  for  $\log h_{gain}$  is  $-0.056$  (M1, CI  $[-0.30, +0.17]$ ),  $+0.049$  (M2,  $[-0.16, +0.28]$ ) and  $-0.203$  (M3,  $[-0.43, -0.00]$ ); for  $\log h_{loss}$ ,  $-0.113$ ,  $-0.404$  (M2,  $[-0.79, -0.01]$ ) and  $-0.161$ . Only two differences exclude zero and both are losses. Poor identification attenuates but does not explain the result: restricting to the 30 well-identified participants lifts baseline  $\log h_{gain}$  only to 0.288.

#### 3.3 Predictive Performance (Choice and RT, by Condition)

*RT.* On the run B RT predictive log-likelihood the three models separate (Fig. 6, Table 4): M2  $-1.492$ , M1  $-1.536$ , M3  $-1.562$  nats per trial, M2 ahead of M1 ( $p = 0.040$ ) and M3 ( $p = 0.004$ ), M1 ahead of M3 ( $p = 0.008$ ). M2 is nominally best in both conditions, has the lowest logRT RMSE (0.754), and is the one ranking here that generalises: on run A it already beat M1 ( $p = 0.005$ ) and M3 ( $p = 0.015$ ). Every model still loses about 0.21 nats per trial between runs, more than any difference between models, because participants speed up by a median of 0.40 s while  $\tau$  and  $a$  are fixed from run A.

*Choice.* Carried across tasks the choice model performs *below* chance ( $\log 0.5 = -0.693$ ): the baseline reaches  $-1.004$  nats per trial, 59.5% balanced accuracy and AUC 0.593, with only 6 of 49 participants above chance, and does better on reward than loss trials ( $-0.897$  against  $-1.111$ ).

*Same-task control.* That collapse is a fact about the two tasks, not the model. Fitting each participant’s choice model on half of run A and predicting the other half – held-out trials, identical estimator,  $\theta_{cert} \equiv 0$  on both sides – gives  $-0.464$  nats per trial and AUC 0.811, with **49 of 49** participants above chance (paired gap  $+0.493$ ,  $p = 1.2 \times 10^{-12}$ ). With Section 3.1, this pins the failure: model and estimator work, and what fails to transfer is a discount rate learned where the safe option is certain.

### 3.4 Value of the RT Modality

(a) *Out-of-sample choice prediction: no reliable effect either way.* Against the baseline the RT models sit at  $\Delta = -0.037$  (M1,  $p = 0.35$ ),  $+0.003$  (M2,  $p = 0.38$ ) and  $+0.025$  (M3,  $p = 0.29$ ) nats per trial, 23 to 26 of 49 participants improved (Fig. 7); balanced accuracy falls slightly in all three and nothing approaches significance. Unlike the RT comparison the direction is not even consistent: for the choices, modelling RT is a wash.

(b) *Identifiability and reliability: RT helps in simulation, not in the data.* Simulation says RT should help substantially, most under M3, and that gain survives a change of task structure (§3.1); on real data none of it appears, and the only two differences excluding zero are losses (§3.2). Effect size reconciles them: the simulation draws  $b$  from ranges implying a strong difficulty effect; the real one is near null. The within-participant slope of  $\log RT$  on  $|\Delta V|$  is  $-0.027$  and the raw pooled slope  $+0.063$ , so even its sign depends on how participant-level RT offsets are handled, and median fitted  $b$  is weakly positive but negative in 20, 11 and 16 of 49 participants, so the RT term injects variance into the shared parameters rather than information. Modelling RT therefore buys an account of response time itself, and under M2 a mapping that replicates across runs, but neither a better account of the choices nor a more stable discount rate.

## 4 DISCUSSION

**We recommend M2, decision uncertainty.** On the quantity the brief designates for separating mappings, the run B RT predictive log-likelihood, M2 is best at  $-1.492$  nats per trial against  $-1.536$  and  $-1.562$ , and the tests separate all three ( $p = 0.040, 0.004, 0.008$ ). It leads in both conditions and is the only ranking here that survives the move to a structurally different run B. Psychologically that favours saturation: what slows a participant is indecision, not value distance. The counterweight is that M2 is the weakest of the three at what RT was meant to buy us: its simulated recovery gain for  $\log h_{gain}$  is nil (RMSE ratio 0.98 against M3’s 0.47) and it most damages the stability of  $\log h_{loss}$  ( $\Delta r = -0.404$ , CI  $[-0.79, -0.01]$ ). We recommend it as the best description of response time, not as an improvement to the discounting model.

**Was modelling RT worthwhile?** Not for the choices, where the three mappings move the choice predictive log-likelihood by at most 0.037 nats per trial in either direction, none significant; and not for the constructs, where the large simulated recovery gains fail to materialise. Model RT if response time is the object of study, not for a better discount rate.

**What we could not establish, and limitations.** We cannot claim any mapping is true: M2 wins by 0.04–0.07 nats per trial against a between-run drop of  $\approx 0.20$  that hits every model equally. Nor can we estimate a transferable discount rate, and that is structural, not a modelling failure.  $h$  recovers from run-B-structure designs ( $r = 0.82\text{--}0.93$ ), and the same-task control predicts held-out run A choices at  $-0.464$  nats per trial with 49 of 49 participants above chance, against  $-1.004$  and 6 of 49 across tasks: model and estimator are sound, transfer is what fails, and §3.2 follows from it. Three limitations bound the rest. The RT model is descriptive, not mechanistic: a shifted log-normal with a value-driven location cannot separate drift from boundary, so a speed–accuracy shift and a change in valuation are confounded in  $b$ , the likeliest reason RT can damage  $h$ .  $\tau$  is unidentified, and the floor applied to run B RTs below run A’s  $\tau$  is worth about 0.05 nats per trial, the same order as the gaps between RT models; M2 is also the least numerically stable model, a caveat we state rather than hide about the model we recommend. Finally the model is deliberately simple: estimation is per participant with no pooling and 19 of 49 are poorly identified, one  $\beta$  is shared across conditions, and  $\Delta V$  is normalised so magnitude effects vanish by construction.

**Next steps.** Fit a genuine sequential-sampling model, DDM or LBA, in which  $\Delta V$  drives drift and a separate boundary carries speed–accuracy (Rodriguez et al., 2014; Dai & Busemeyer, 2014): that tests whether RT fails to sharpen the discount rates because the signal is absent or because our functional form is wrong. Second, stop assuming one  $h$  transfers: estimate a rate per task structure and ask how the two relate.

## FIGURES

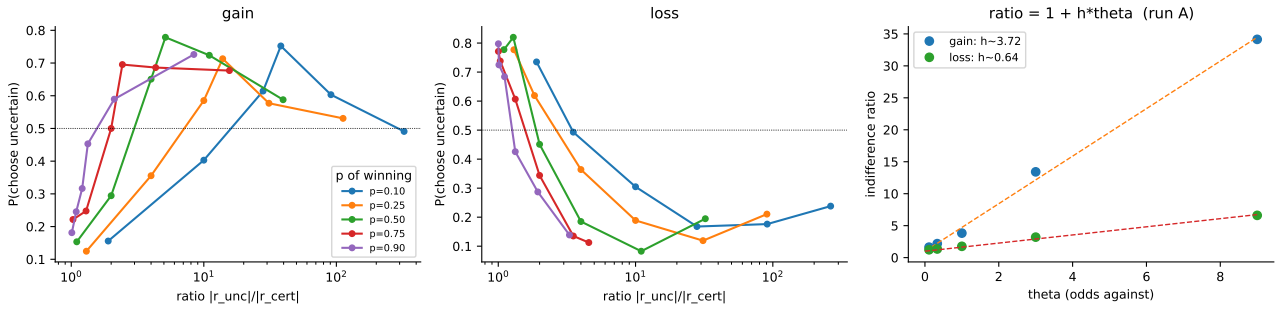


Figure 1: **Model-free data summary.** Left and middle: proportion of uncertain choices against the magnitude ratio  $|r_{unc}|/|r_{cert}|$ , binned into sextiles, separately for each outcome probability  $p$ , in the reward (left) and loss (middle) conditions. Right: indifference ratios estimated at each  $p$  against the odds against  $\theta$ , on run A only, where the smaller option is certain. The hyperboloid predicts  $r_{unc}/r_{cert} = 1 + h\theta$  (dashed lines), giving model-free  $h \approx 3.72$  for gains and  $h \approx 0.64$  for losses.

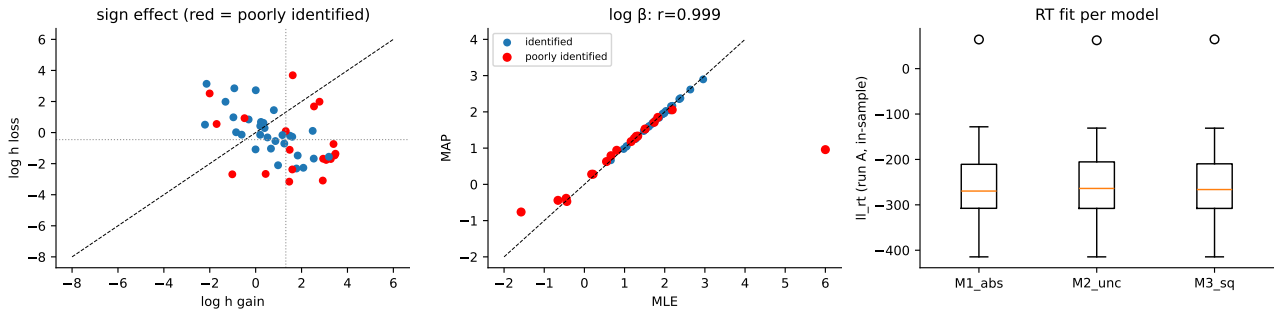


Figure 2: **Inference diagnostics (run A).** Left: fitted  $\log h_{gain}$  against  $\log h_{loss}$  per participant, with those flagged as poorly identified by the curvature diagnostic in red and the model-free rates marked by dotted lines; the mass below the diagonal is the reflection effect. Middle: MAP against MLE estimates of  $\log \beta$ , showing the priors do not move estimates where the likelihood has curvature. Right: in-sample run A RT log-likelihood per model.

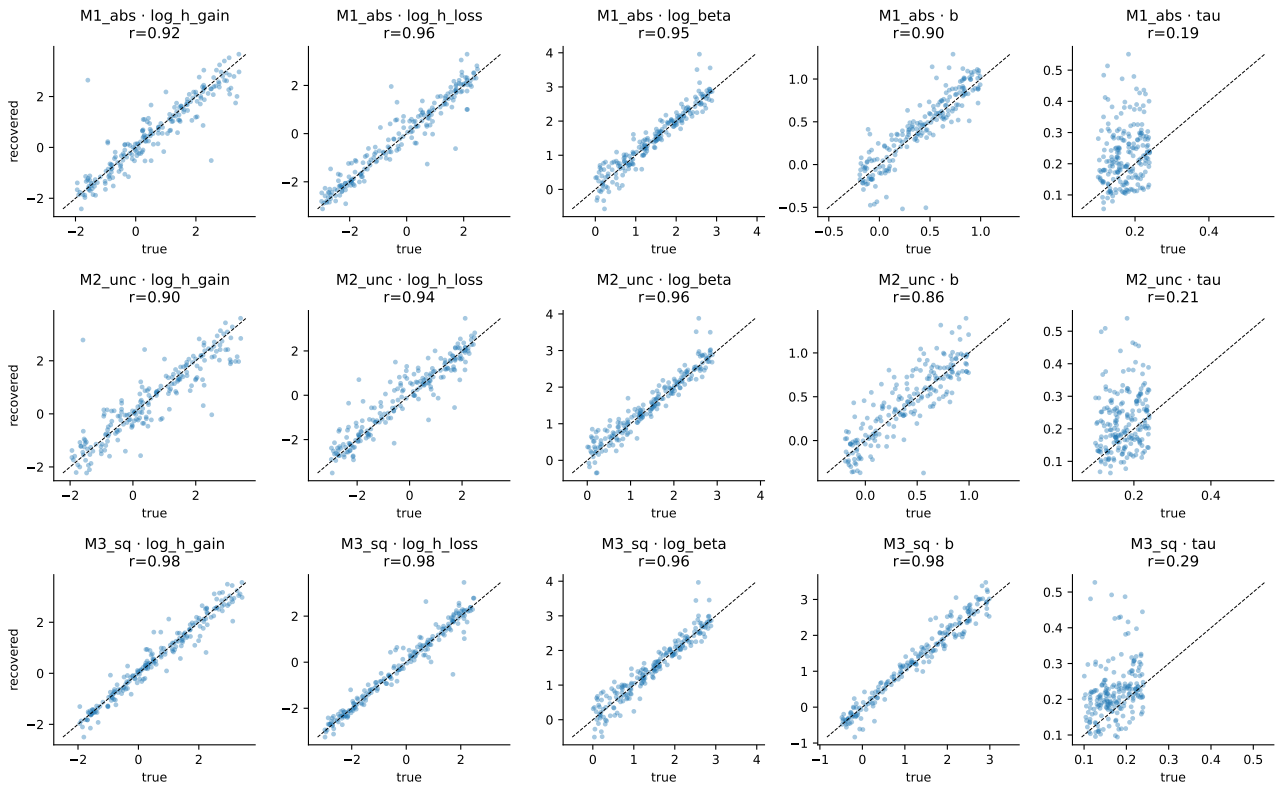


Figure 3: **Parameter recovery.** True against recovered values over 200 simulated participants per model (rows: M1, M2, M3) for the two discount rates, choice sensitivity, the difficulty slope  $b$  and non-decision time  $\tau$ , on run A designs. Dashed line is identity. All choice parameters recover well;  $\tau$  does not ( $r = 0.19$ – $0.29$ ). Repeating the study on run B designs gives  $r = 0.82$ – $0.93$  for the discount rates (§3.1).

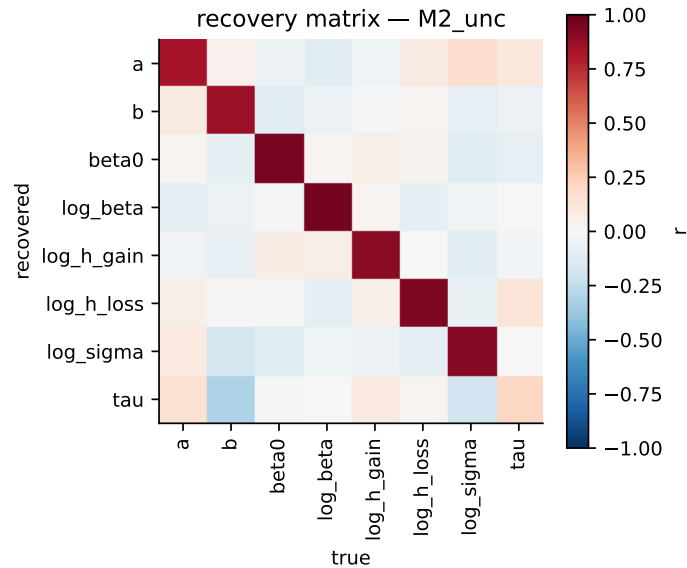


Figure 4: **Recovery matrix (M2).** Correlation between each true parameter (columns) and each recovered parameter (rows). Off-diagonal structure indicates trade-offs; the only appreciable one is  $\tau$  against  $b$  and  $a$ .

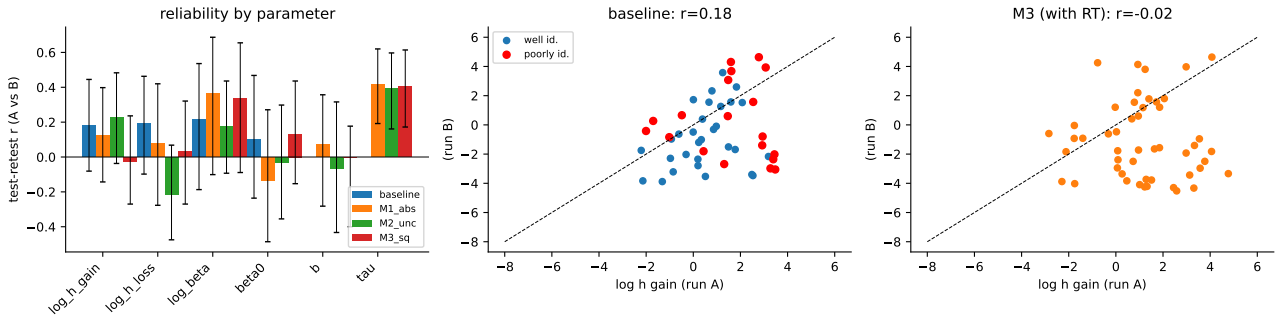


Figure 5: **Cross-context reliability (run A against run B).** Left: reliability per parameter and model with bootstrap 95% CIs; note that every CI for a discounting parameter crosses zero, while the RT level  $a$ ,  $\log \sigma$  and  $\tau$  are stable. Middle and right: run A against run B estimates of  $\log h_{gain}$  for the choice-only baseline (red marks participants flagged as poorly identified by the curvature diagnostic) and for M3, the mapping whose reduction in stability is the one large enough to exclude zero.

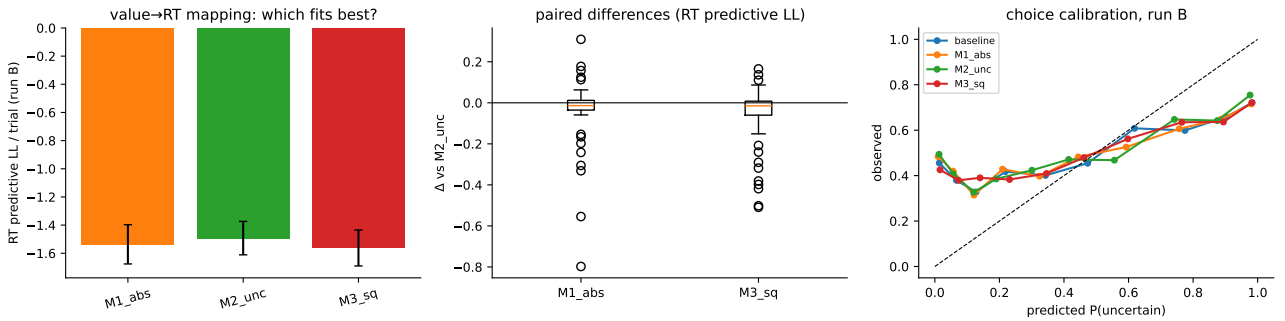


Figure 6: **Out-of-sample prediction (run B).** Left: RT predictive log-likelihood per trial by model (mean  $\pm$  SEM across participants), the common quantity on which the RT models are compared. Middle: paired per-participant differences against the nominally best model. Right: calibration of predicted against observed choice probability, decile-binned.



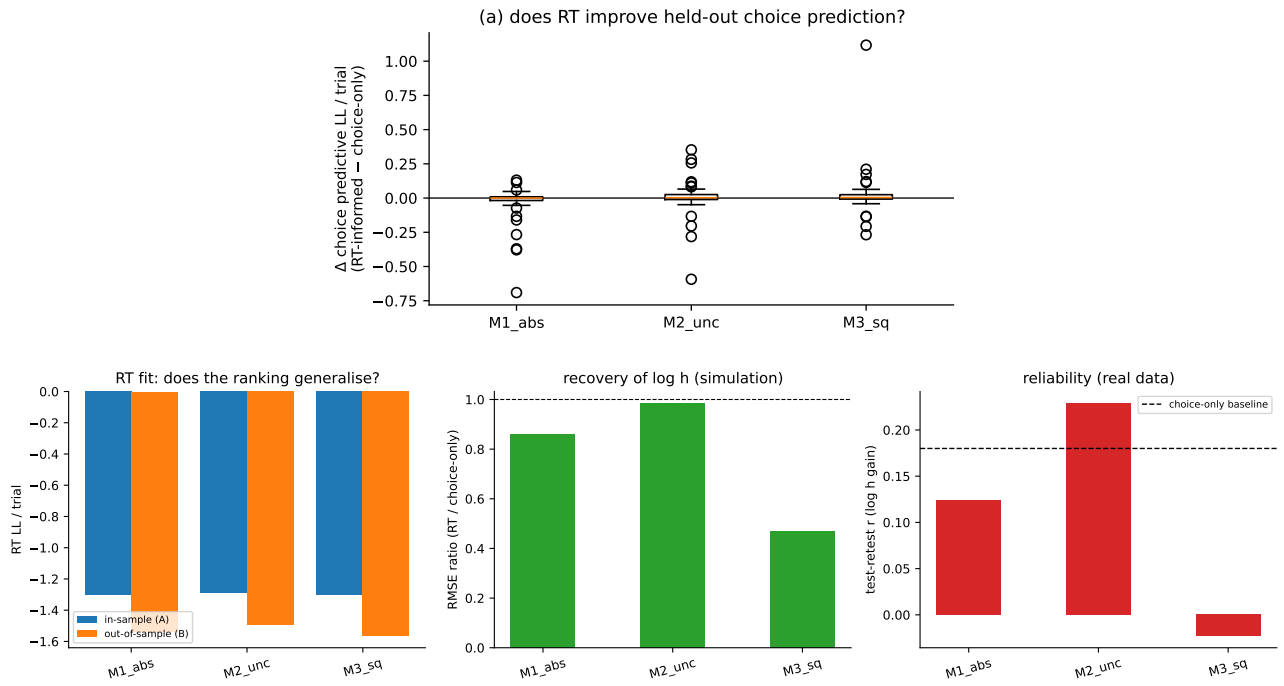


Figure 7: **Value of the RT modality.** Top: per-participant difference in run B choice predictive log-likelihood between each RT-informed fit and the choice-only baseline; values below zero mean RT hurts choice prediction. Bottom left: in-sample against out-of-sample RT fit, showing both that the run A ranking does not generalise and how much every model loses between sessions. Bottom middle: recovery RMSE ratio against the choice-only fit, where below the dashed line means RT improves recovery. Bottom right: test–retest reliability of  $\log h_{\text{gain}}$  per RT model against the choice-only baseline (dashed).

## TABLES

Table 1: Priors on the fitted (transformed) scale, and their rationale.

| Parameter              | Prior                                                | Rationale                                                                                                                                                                                                                 |
|------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| log_h_gain, log_h_loss | $\mathcal{N}(0, 2^2)$                                | Centred at $h = 1$ , with 95% mass over $h \in [0.02, 55]$ , far wider than the model-free estimates (3.72, 0.64), so the data dominate.                                                                                  |
| log_beta               | $\mathcal{N}(1, 1.5^2)$                              | Covers $\beta \in [0.14, 52]$ ; excludes only numerically degenerate values.                                                                                                                                              |
| beta0                  | $\mathcal{N}(0, 2^2)$                                | No stance on the direction of bias.                                                                                                                                                                                       |
| a                      | $\mathcal{N}(0.5, 1^2)$                              | Centred near $\log(\text{median RT} - \tau) \approx 0.6$ .                                                                                                                                                                |
| b                      | $\mathcal{N}(0, 1^2)$ ; $\mathcal{N}(0, 3^2)$ for M3 | Centred at zero and symmetric, so a positive difficulty effect is not assumed. Widened for M3 because $\Delta V^2$ is on a smaller scale, which keeps regularisation comparable in effect rather than in raw coefficient. |
| log_sigma              | $\mathcal{N}(-0.5, 0.5^2)$                           | Matches the observed spread of log RT.                                                                                                                                                                                    |
| tau_raw                | $\mathcal{N}(0, 1.5^2)$                              | Nearly flat over the admissible interval $(0, \min RT)$ .                                                                                                                                                                 |

Table 2: Parameter recovery over 200 simulated participants per model.  $r$  and RMSE are between true and recovered values from the full choice + RT fit; the RMSE ratio is (joint fit)/(choice-only fit), so values below 1 mean RT improves recovery.  $b$ 's RMSE is not comparable across models because the predictors are on different scales.

| Parameter      | M1 (linear)  |       |       | M2 (uncertainty) |       |       | M3 (quadratic) |       |              |
|----------------|--------------|-------|-------|------------------|-------|-------|----------------|-------|--------------|
|                | $r$          | RMSE  | ratio | $r$              | RMSE  | ratio | $r$            | RMSE  | ratio        |
| log $h_{gain}$ | 0.925        | 0.569 | 0.862 | 0.900            | 0.649 | 0.983 | <b>0.977</b>   | 0.319 | <b>0.471</b> |
| log $h_{loss}$ | 0.958        | 0.486 | 0.719 | 0.936            | 0.598 | 0.883 | <b>0.978</b>   | 0.357 | <b>0.490</b> |
| log $\beta$    | 0.955        | 0.266 | 0.964 | 0.958            | 0.253 | 0.917 | 0.956          | 0.263 | 0.923        |
| $\beta_0$      | 0.954        | 0.371 | 0.869 | 0.952            | 0.382 | 0.895 | 0.963          | 0.326 | 0.760        |
| $a$            | 0.884        | 0.107 | –     | 0.825            | 0.145 | –     | 0.928          | 0.086 | –            |
| $b$            | 0.897        | 0.184 | –     | 0.860            | 0.210 | –     | 0.978          | 0.238 | –            |
| log $\sigma$   | 0.917        | 0.097 | –     | 0.914            | 0.098 | –     | 0.935          | 0.088 | –            |
| $\tau$         | <b>0.187</b> | 0.110 | –     | <b>0.207</b>     | 0.103 | –     | <b>0.287</b>   | 0.089 | –            |

Table 3: Cross-context reliability (Pearson  $r$  between run A and run B estimates,  $n = 49$ ). Because run A is sure-vs-risky and run B risky-vs-risky, this measures generalisation of a construct across decision structures rather than classical test–retest. Bracketed values are bootstrap 95% CIs (2000 resamples; for the  $\Delta$  columns the resample is applied to both models, preserving the pairing). Dashes mark parameters absent from the baseline; starred differences exclude zero.

| Parameter      | baseline                   | M1           | M2           | M3           | $\Delta M1$ | $\Delta M2$ | $\Delta M3$ |
|----------------|----------------------------|--------------|--------------|--------------|-------------|-------------|-------------|
| log $h_{gain}$ | 0.180 [−0.08, 0.45]        | 0.124        | <b>0.229</b> | −0.023       | −0.056      | +0.049      | −0.203*     |
| log $h_{loss}$ | <b>0.191</b> [−0.10, 0.46] | 0.077        | −0.213       | 0.030        | −0.113      | −0.404*     | −0.161      |
| log $\beta$    | 0.215 [−0.19, 0.54]        | <b>0.365</b> | 0.175        | 0.336        | +0.150      | −0.040      | +0.121      |
| $\beta_0$      | 0.103 [−0.24, 0.47]        | −0.132       | −0.033       | <b>0.132</b> | −0.235      | −0.136      | +0.029      |
| $a$            | –                          | <b>0.754</b> | 0.493        | 0.747        | –           | –           | –           |
| $b$            | –                          | 0.070        | −0.068       | −0.002       | –           | –           | –           |
| log $\sigma$   | –                          | 0.660        | <b>0.668</b> | 0.662        | –           | –           | –           |
| $\tau$         | –                          | <b>0.418</b> | 0.392        | 0.407        | –           | –           | –           |

Table 4: Out-of-sample performance on run B, participant means. Run A parameters are applied unchanged. RT models are compared with each other on  $LL_{RT}$  only; RT-informed fits are compared with the baseline on  $LL_{choice}$  and balanced accuracy only. All log-likelihoods are per trial, and chance for choice is  $-0.693$ . The final row is the same-task control of §3.3: the identical pipeline fitted on half of run A and evaluated on the other half, which is the only row in the table that is not a cross-task extrapolation.

| Model                                                          | Condition | $LL_{choice}$              | bal. acc.    | AUC          | $LL_{RT}$                  | RMSE logRT   |
|----------------------------------------------------------------|-----------|----------------------------|--------------|--------------|----------------------------|--------------|
| baseline                                                       | all       | $-1.004$                   | <b>0.595</b> | <b>0.593</b> | –                          | –            |
|                                                                | reward    | $-0.897$                   | 0.585        | 0.637        | –                          | –            |
|                                                                | loss      | $-1.111$                   | 0.569        | 0.592        | –                          | –            |
| M1                                                             | all       | $-1.041$                   | 0.587        | 0.589        | $-1.536$                   | 0.773        |
|                                                                | reward    | $-0.885$                   | 0.597        | 0.652        | $-1.629$                   | 0.800        |
|                                                                | loss      | $-1.196$                   | 0.546        | 0.563        | $-1.444$                   | 0.730        |
| M2                                                             | all       | $-1.001$                   | 0.588        | 0.585        | <b><math>-1.492</math></b> | <b>0.754</b> |
|                                                                | reward    | <b><math>-0.826</math></b> | <b>0.610</b> | <b>0.659</b> | <b><math>-1.552</math></b> | <b>0.774</b> |
|                                                                | loss      | $-1.176$                   | 0.538        | 0.542        | <b><math>-1.433</math></b> | <b>0.718</b> |
| M3                                                             | all       | <b><math>-0.979</math></b> | 0.590        | 0.591        | $-1.562$                   | 0.790        |
|                                                                | reward    | $-0.888$                   | 0.599        | 0.627        | $-1.616$                   | 0.806        |
|                                                                | loss      | <b><math>-1.071</math></b> | 0.551        | 0.582        | $-1.508$                   | 0.755        |
| <i>same-task control</i> (half run A $\rightarrow$ half run A) |           | <b><math>-0.464</math></b> | –            | <b>0.811</b> | –                          | –            |

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